



AI-Assistant Chatbot for Real-Time Information of Finnish Railway

Cornelius Niese, Paul Horn, Nina-Sophie Zink

Abstract

Keywords

Keyword1, Keyword2, Keyword3 ...

Advisors: Aleš Žagar

Introduction

Features:

- give departure and arrival times of trains (based on train identifier or start and end station + time)
- give travel routes from one station to another
- give delay time of specific trains
- give prices for tickets

Potential datasets:

- Digitraffic Open API (real-time, main source)
- Fintraffic Train Departures: <https://junalahdot.fintraffic.fi/?lang=3> (real-time)
- Network Statement of Finnish Transport Infrastructure Agency: PDF file with detailed information on all tracks and stations

Related Work

- Sharma et al. [1]
- Xu et al. [2]
- Rafee et al. [3]
- Dharani et al. [4]
- Fantin et al. [5]
- Banu et al. [6]

- Prathamesh et al. [7]
- BetterBahn by Lukas Weihrauch (<https://github.com/BetterBahn/betterbahn?tab=readme-ov-file>)
- Wang et al. [8]

References

- [1] Sanat Sharma, David Seunghyun Yoon, Franck Deroncourt, Dewang Sultania, Karishma Bagga, Mengjiao Zhang, Trung Bui, and Varun Kotte. Retrieval Augmented Generation for Domain-specific Question Answering, May 2024. arXiv:2404.14760 [cs].
- [2] Haowen Xu, Jinghui Yuan, Anye Zhou, Guanhao Xu, Wan Li, Xuegang Ban, and Xinyue Ye. GenAI-powered Multi-Agent Paradigm for Smart Urban Mobility: Opportunities and Challenges for Integrating Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) with Intelligent Transportation Systems, September 2024. arXiv:2409.00494 [cs].
- [3] Bus Enquiry Chatbot for Transportation System with Voice Support and Emergency Assistance Using NLP – IJSREM.
- [4] M. Dharani, J.V.S.L. Jyostna, E. Sucharitha, R. Likitha, and Suneetha Manne. Interactive Transport Enquiry with AI Chatbot. In *2020 4th International Conference on Intelligent Computing and Control Systems (ICICCS)*, pages 1271–1276, May 2020.
- [5] Luca Fantin, Marco Antonelli, Margherita Cesetti, Daniele Irto, Bruno Zamengo, and Francesco Silvestri. Design and

testing of an agent chatbot supporting decision making with public transport data, May 2025. arXiv:2505.22698 [cs].

- [6] Sajitha Banu, Narendra Chari, and Lavanya Lavanya. Railway assistance chatbot using ann and nlp. *INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*, 09(04):1–9, 2025.
- [7] Lohote Prathamesh, Belhaware Akshay, Barve Omkar, Doke Sanket, and Gholap P. S. An intelligent ai for transportation system. *International Journal of Advanced Research in Science, Communication and Technology*, pages 337–341, 2022.
- [8] Bingzhang Wang, Zhiyu Cai, Muhammad Monjurul Karim, Chenxi Liu, and Yinhai Wang. Traffic Performance GPT (TP-GPT): Real-Time Data Informed Intelligent ChatBot for Transportation Surveillance and Management. In *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, pages 460–467, September 2024. ISSN: 2153-0017.